Exercise 13: SAT Final Testbench

Objective: *Complete a testbench by implementing checkers identified in a verification plan.*

Task: *Implementation and integration of the remaining design checkers for the Saturation Filter testbench.*

The files for this exercise are in `<ROOT>/lecture08/exercises/E13_sat_final_testbench/`. This directory contains the RTL and testbench files. In the following sections, file names are relative to `tb/`.

In Exercise 12, you created a verification plan for the *Saturation Filter*. The testbench is missing two checkers: a propagation checker for RS/03 and an overflow checker for RS/05. Implement and integrate both design checks in this exercise.

Part 1: Checkers

In `sat_checkers.py`, implement a class named `sat_checkers`.

- The constructor shall accept and store the producer and consumer virtual interfaces as `producer_vif` and `consumer_vif`, and the overflow signal as `overflow_signal`.
- The checker shall have a `passed` member variable, initially set to `True`. Set `passed` to `False` if either design check fails.
- Implement an assertion, which checks **RS/03** for the `valid` signals.
- Implement an assertion, which checks **RS/05**.
- Implement a `check_assertions()` method that starts the propagation-checker.

Part 2: Testbench Integration

Integrate the design checkers into `sat_tb_base_test.py`.

- Import `sat_checkers` from `sat_checkers.py`.
- Add a `design_checkers` handle to the test class constructor.
- In `build_phase`, create the checker with the producer and consumer interfaces and the overflow signal from the DUT.
- In `run_phase`, start the design checkers.
- In `report_phase`, assert the `passed` attribute so that a checker violation fails the test.